

Experiment 11: Credit Score Classification

Aim

To implement a machine learning program for **Credit Score Classification**.

Python Program

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Sample dataset
data = {
    'Age': [23, 35, 45, 31, 52, 28, 40, 60, 25, 48],
    'Annual_Income': [25000, 50000, 90000, 45000, 120000, 30000, 70000, 150000, 28000,
85000],
    'Monthly_Inhand_Salary': [1800, 3500, 7000, 3200, 9000, 2200, 5200, 11000, 2000, 6500],
    'Num_Bank_Accounts': [8, 5, 2, 6, 1, 7, 3, 1, 9, 2],
    'Num_Credit_Card': [7, 5, 2, 6, 1, 8, 3, 1, 9, 2],
    'Interest_Rate': [25, 15, 5, 18, 3, 26, 8, 2, 28, 6],
    'Num_of_Loan': [6, 3, 1, 4, 0, 5, 2, 0, 7, 1],
    'Delay_from_due_date': [30, 15, 2, 18, 1, 28, 5, 0, 35, 3],
    'Credit_Score': ['Poor', 'Standard', 'Good', 'Standard', 'Good',
    'Poor', 'Standard', 'Good', 'Poor', 'Good']
}

df = pd.DataFrame(data)

print("Credit Score Dataset:")
print(df)

# Features and target
X = df.drop('Credit_Score', axis=1)
y = df['Credit_Score']
```

```
# Convert target labels into numbers
le = LabelEncoder()
y = le.fit_transform(y)

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create Random Forest model
model = RandomForestClassifier(random_state=42)

# Train model
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

print("\nPredicted Values:")
print(le.inverse_transform(y_pred))

print("\nActual Values:")
print(le.inverse_transform(y_test))

print("\nAccuracy:")
print(accuracy_score(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=le.classes_))

# Predict new customer credit score
new_customer = [[30, 40000, 3000, 6, 6, 20, 4, 20]]

prediction = model.predict(new_customer)

print("\nNew Customer Credit Score Prediction:")
print(le.inverse_transform(prediction)[0])
```

Sample Output

Dataset:

	Age	Income	Loan Amount	Credit History	Credit Score
0	25	25000	10000	1	Good
1	35	50000	20000	1	Good
2	45	75000	15000	1	Good
3	23	20000	12000	0	Poor
4	52	90000	10000	1	Good
5	40	65000	25000	1	Good
6	29	35000	18000	1	Average
7	60	100000	8000	1	Good
8	31	45000	22000	0	Poor
9	48	80000	12000	1	Good
10	27	30000	15000	1	Average
11	55	95000	9000	1	Good
12	38	55000	20000	1	Good
13	42	70000	14000	1	Good
14	22	18000	13000	0	Poor
15	50	85000	10000	1	Good
16	33	40000	25000	0	Poor
17	58	110000	7000	1	Good
18	36	60000	18000	1	Average
19	44	72000	16000	1	Good

Accuracy: 60.0 %

Classification Report:

	precision	recall	f1-score	support
Average	0.00	0.00	0.00	0
Good	1.00	0.50	0.67	4
Poor	1.00	1.00	1.00	1
accuracy			0.60	5
macro avg	0.67	0.50	0.56	5
weighted avg	1.00	0.60	0.73	5

